

OSCAR RAFAEL LAM HO

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SUMMARY

Data engineer (B.S. Computer Science) building end-to-end ELT pipelines on Snowflake, dbt, and AWS S3, with Dagster orchestration and cold-path streaming ingestion via Spark Structured Streaming. Strong in dimensional modeling, data-quality testing, and turning large public datasets into decision-ready analytics. Targeting Junior Data Engineer / Analytics Engineer roles.

PROJECT EXPERIENCE

NYC 311 Service Equity Mart — <https://nyc-311-service-equity.streamlit.app/>

Apr 2026 – May 2026

Snowflake · dbt · AWS S3 · Python · Streamlit · GitHub Actions · Dagster

- Quantified a persistent 1.5–2x resolution-time gap between top- and bottom-income census quintiles across 1.7M volume-filtered NYC 311 requests, applying a per-tract sample-size floor to suppress statistically unstable results
- Engineered an incremental Socrata→S3→Snowflake ELT pipeline ingesting 21M+ records; guaranteed idempotent re-runs via staging dedup (ROW_NUMBER over unique_key) and a 48-hour watermark buffer, preventing duplicate-row drift on overlapping daily loads
- Geocoded 1.7M filtered requests to 2,168 census tracts with geopandas; orchestrated end-to-end in Dagster with source-freshness monitoring and CI validation via GitHub Actions
- Modeled a layered dbt project (staging→intermediate→marts) with 40+ tests across 9 models — including a custom percentile-ordering invariant ($P50 \leq P75 \leq P90$) — joining civic data to U.S. Census demographics, packaged in a live Streamlit dashboard for non-technical civic stakeholders

MTA Subway Reliability Tracker

Apr 2026 – May 2026

Kafka (Redpanda) · Spark Structured Streaming · AWS S3 · Snowflake · dbt · GitHub Actions

- Diagnosed and corrected three data-quality defects (a 134x history inflation, a 4-hour UTC/EDT offset, and a GTFS past-midnight bias) to reach 15-second median schedule-vs-feed agreement (90% of L-train trips within 30s) before extrapolating system-wide with documented limitations
- Built a cold-path streaming pipeline (Redpanda → Spark Structured Streaming → partitioned S3 Parquet → Snowflake COPY INTO) ingesting MTA GTFS-Realtime feeds; chose cold-path-only after finding delay_seconds populated for just 1 of 30 lines, forcing a schedule-derived delay computation
- Modeled on-time rates and delay percentiles across all 30 subway lines in a layered dbt project, applying a sample-size floor (HAVING COUNT >= 30) to suppress single-arrival noise buckets

Supply Chain Intelligence Engine

Feb 2026 – Present

Python · PySpark · DuckDB · dbt · Great Expectations · LightGBM · Plotly/Dash · GitHub Actions

- Designed a version-controlled YAML data catalog decoupling source curation from pipeline logic, enabling commodity-scope changes across 30+ USDA/FRED sources without code redeployments
- Gated data quality at the curation boundary with Great Expectations across a weekly automated feature-store refresh on a GitHub Actions schedule
- Built an automated weekly feature-store refresh (22 engineered lag, rolling, and volatility features) on a GitHub Actions schedule, feeding a 13-week quantile price forecast (p10/p50/p90) with SHAP attribution across 10+ commodities
- Delivered a 7-tab Dash dashboard exposing a 4-signal composite supply-risk score and a linear-programming 52-week procurement optimizer

TECHNICAL SKILLS

Languages & Data: Python, SQL, Bash, Spark (Structured Streaming), Pandas, dbt, Great Expectations

Cloud & Warehousing: Snowflake, AWS (S3, IAM, Athena)

Orchestration & Streaming: GitHub Actions CI/CD, Kafka / Redpanda, Spark Structured Streaming, Airflow, Dagster

Modeling & Quality: Dimensional / star-schema modeling, ELT, incremental models, data-quality testing, idempotency, partitioning

Dev & Apps: Docker, docker-compose, Git, Streamlit, Plotly/Dash, FastAPI

WORK EXPERIENCE

Luckin Coffee | Assistant Manager — New York, NY

Dec 2025 – Apr 2026

- Tracked hourly cup counts and customer-satisfaction metrics across 100+ daily drink variations; trained ~2 new hires per week with structured onboarding plans
- Managed daily and weekly inventory counts for 50+ raw products to minimize waste and maintain supply continuity

EDUCATION

CUNY Brooklyn College | B.S. Computer Science — GPA 3.83 / 4.0

Feb 2025

CERTIFICATIONS

AWS Certified Data Engineer – Associate (in progress, exam Jun 2026) • dbt Analytics Engineer Certification (planned Jul 2026)